

Classical Cryptography

Classical Cryptography:

“Classical Cryptography was confined to the art of designing and breaking encryption schemes (or secrecy codes)”

Modern Cryptography:

“Modern Cryptography is concerned with the rigorous analysis of any system which should withstand malicious attempts to abuse it.”

It **emphasizes two aspects** of the transition from classical to modern cryptography:

- 1) the **widening of scope** from one specific task to an utmost wide general class of tasks
- 2) the move from an engineering-art which strives **on ad-hoc tricks to a scientific discipline** based on rigorous approaches and techniques.

Definitions

Plaintext

- An original message

Ciphertext

- The coded message

Enciphering /encryption

- The process of converting from plaintext to ciphertext

Deciphering/decryption

- Restoring the plaintext from the ciphertext

Cryptography

- The area of study of the many schemes used for encryption

Cryptographic system /cipher

- A scheme

Cryptanalysis

- Techniques used for deciphering a message without any knowledge of the enciphering details

Cryptology

- The areas of cryptography and cryptanalysis

Encryption Models

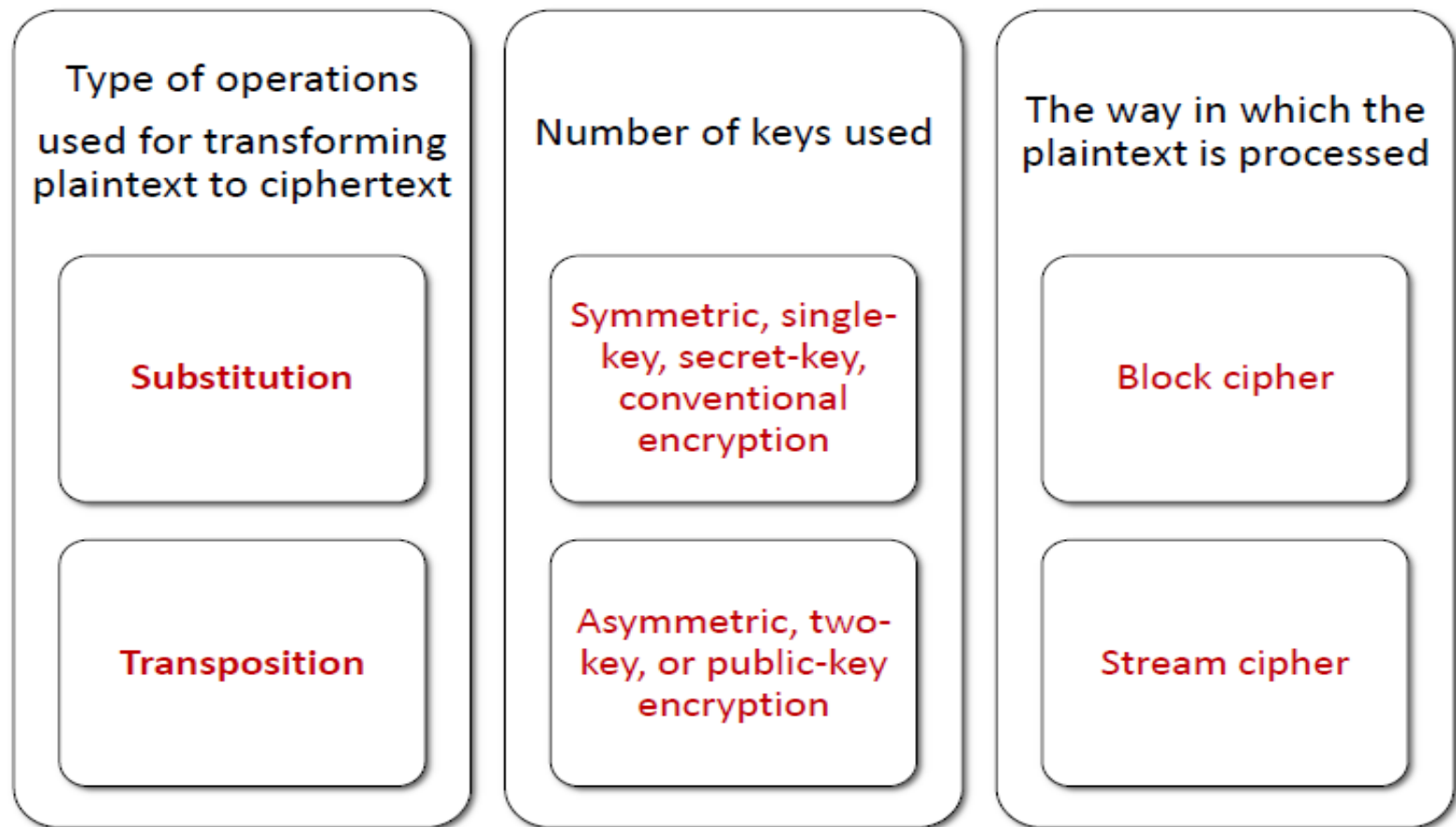
Symmetric Encryption



Asymmetric Encryption



Cryptographic Systems



Cryptanalysis and Brute-Force Attack

Two general approaches to attacking a conventional encryption scheme:

Cryptanalysis

Attack relies on the nature of the algorithm **plus** some knowledge of the general characteristics of the plaintext

Attack **exploits the characteristics of the algorithm** to attempt to deduce a specific plaintext or to deduce the key being used

Brute-force attack

Attacker **tries every possible key** on a piece of cipher-text until an intelligible translation into plaintext is obtained

On average, half of all **possible keys must be tried** to achieve success

Encryption Scheme Security

Unconditionally secure

No matter how much time an opponent has, **it is impossible for him or her to decrypt the ciphertext** simply because the required information is not there

Computationally secure

The **cost of breaking the cipher** exceeds the value of the encrypted information

The **time required to break** the cipher exceeds the useful lifetime of the information



Caesar Cipher Algorithm

Can define transformation as:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Algorithm can be expressed as:

$$c = E(3, p) = (p + 3) \bmod (26)$$

A shift may be of any amount, so that the general Caesar algorithm is:

$$C = E(k, p) = (p + k) \bmod 26$$

where k takes on a value in the range 1 to 25;

The decryption algorithm is simply:

$$p = D(k, C) = (C - k) \bmod 26$$



Brute-Force Cryptanalysis of Caesar Cipher

If it is known that a given ciphertext is a Caesar cipher, then a brute-force cryptanalysis is easily performed:

- simply try all the 25 possible keys

Three important characteristics of this problem enabled us to use a brute-force cryptanalysis:

- 1) The encryption and decryption algorithms are known.
- 2) There are only 25 keys to try.
- 3) The language of the plaintext is known and easily recognizable.

So, with only 25 possible keys, the Caesar cipher is far from secure!

	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
KEY						
1	oggv	og	chvgt	vjg	vqic	rectva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozgsx
5	kccr	kc	ydrpc	rfe	rmey	nyprw
6	jbbq	jb	xcqbo	qeb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	ojbv	kvmtot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlg
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fqhjo
14	btti	bt	putg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgre	gur	gbtn	cnegl
17	yqqf	yq	mrfqd	ftq	fasm	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnc	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzcx	znk	zumg	vgxze
24	rjyy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxc

Shift Cipher

- $Z_{26} = \{0, 1, 2, \dots, 24, 25\}$
- $\mathcal{P} = \mathcal{C} = \mathcal{K} = Z_{26}$ $\oplus$ $\oplus$
- For $k \in \mathcal{K}$,
 $e_k(x) = (x + k) \bmod 26$ for $x \in \mathcal{P}$
 $d_k(y) = (y - k) \bmod 26$ for $y \in \mathcal{C}$
- Caesar Cipher is a particular case (for $k = 3$)

Substitution Cipher

- $\mathcal{P} = \mathcal{C}$ = set of 26-letter English alphabet
$$\mathcal{P} = \{a, b, c, \dots, y, z\}$$
$$\mathcal{C} = \{A, B, C, \dots, Y, Z\}$$
- $\mathcal{K}$ = set of all possible permutations of 26 alphabetic characters.
- For each permutation $\phi \in \mathcal{K}$,
$$e_{\phi}(x) = \phi(x) \text{ for } x \in \mathcal{P}$$
$$d_{\phi}(y) = \phi^{-1}(y) \text{ for } y \in \mathcal{C}, \text{ where } \phi^{-1} \text{ is the inverse permutation of } \phi.$$

Example

- Encryption function is the permutation ϕ :

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
X	N	Y	A	H	P	O	G	Z	Q	W	B	T	S	F	L
q	r	s	t	u	v	w	x	y	z						
R	C	V	M	U	E	K	J	D	I						

- Decryption function is the inverse permutation ϕ^{-1} :

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
d	l	r	y	v	o	h	e	z	x	w	p	t	b	g	f
Q	R	S	T	U	V	W	X	Y	Z						
j	q	n	m	u	s	k	a	c	i						

Frequency Analysis

- Suppose we have a long ciphertext, the challenge is to decipher it
- Let us know the text is in English and has been encrypted using a monoalphabetic substitution cipher
- searching all possible keys is impractical as the keyspace is of size $26!$

- In English, *e* is the most common letter, followed by *t*, then *a*, and so on, as shown in the Figure 3

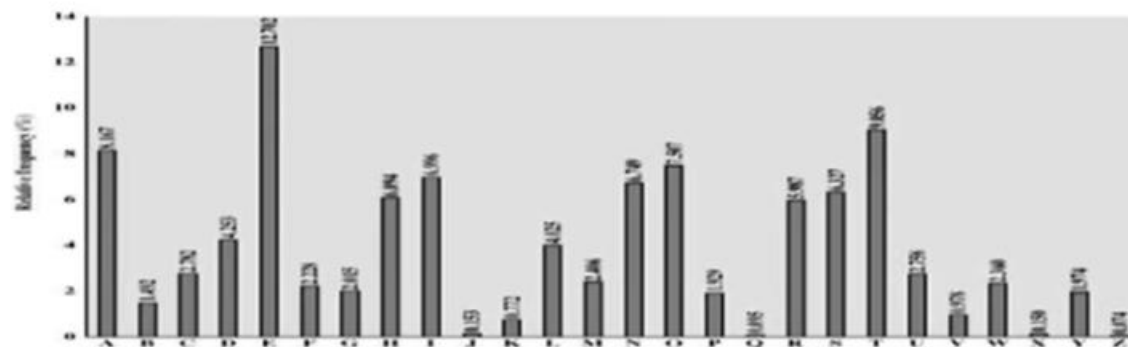


Figure 3: Relative Frequency of letters in the English Language.

- examine the ciphertext in question, and work out the frequency of each letter
- if most common letter in the ciphertext is, for example, J then it would seem likely that this is a substitution for e
- if the second most common letter in the ciphertext is P , then this is probably a substitution for t , and so on
- however, regularities of the language may be exploited, e.g. relative frequency
- frequency analysis requires logical thinking, intuition, flexibility and guesswork

- **Monoalphabetic Cipher:** Each alphabetic character is mapped to a unique alphabetic character
- We use arbitrary monoalphabetic substitution, so there are $26!$ or $4 \times 10^{26} \approx 2^{88}$ possible permutations, which is a very large number. Thus brute force is infeasible.
- However we will see later that a Substitution Cipher is insecure against frequency analysis.

YIFQFMZRWQFYVECFMDZPCVMRZWNMDZVEJB'TXCDDUMJ
 NDI FEFMDZCDMQZKCEYFCJMYRNCWJCSZREXCHZUNMXZ
 NZUCDRJXYYSMRTMEYIFZWDYVZVYFZUMRZCRWNZDZJJ
 XZWGCHSMRNMDHNCMFQCHZJMXJZWIEJYUCFWDJNZDIR

letter	frequency	letter	frequency
<i>A</i>	0	<i>N</i>	9
<i>B</i>	1	<i>O</i>	0
<i>C</i>	15	<i>P</i>	1
<i>D</i>	13	<i>Q</i>	4
<i>E</i>	7	<i>R</i>	10
<i>F</i>	11	<i>S</i>	3
<i>G</i>	1	<i>T</i>	2
<i>H</i>	4	<i>U</i>	5
<i>I</i>	5	<i>V</i>	5
<i>J</i>	11	<i>W</i>	8
<i>K</i>	1	<i>X</i>	6
<i>L</i>	0	<i>Y</i>	10
<i>M</i>	16	<i>Z</i>	20

-----end-----e-----ned---e-----
YIFQFMZRWQFYVECFMDZPCVMRZWNMDZVEJBTXCDDUMJ

-----e-----e-----n--d---en---e---e
NDIFEFMZCDMQZKCEYFCJMYRNCWJCSZREXCHZUNMXZ

-e---n-----n-----ed---e---e--ne-nd-e-e--
NZUCDRJXYYSMRTMEYIFZWVYVZVYFZUMRZCRWNZDZJJ

-ed-----n-----e-----ed-----d---e--n
XZWGCHSMRNMDHNCMFQCHZJMXJZWIEJYUCFWDJNZDIR

o-r-riend-ro--arise-a-inedhise--t---ass-it
YIFQFMZRWQFYVECFMDZPCVMRZWNMDZVEJBTXCDDUMJ

hs-r-riseasi-e-a-orationhadta-en--ace-hi-e
NDIFEFMZCDMQZKCEYFCJMYRNCWJCSZREXCHZUNMXZ

he-asnt-oo-in-i-o-redso-e-ore-ineandhesett
NZUCDRJXYYSMRTMEYIFZWDYVZVYFZUMRZCRWNZDZJJ

-ed-ac-inhischair-aceti-ted--to-ardsthes-n
XZWGCHSMRNMDHNCMFQCHZJMXJZWIEJYUCFWDJNZDIR

Our friend from Paris examined his empty glass with surprise, as if evaporation had taken place while he wasn't looking. I poured some more wine and he settled back in his chair, face tilted up towards the sun.²



Vigenère Cipher

- **Polyalphabetic cipher:** use different monoalphabetic substitutions while moving through the plaintext.
- Let m be a positive integer
- $\mathcal{P} = \mathcal{C} = \mathcal{K} = (Z_{26})^m$
- For $k = (k_1, k_2, \dots, k_m) \in \mathcal{K}$,
$$e_k(x_1, x_2, \dots, x_m) = (x_1 + k_1, x_2 + k_2, \dots, x_m + k_m)$$
$$d_k(y_1, y_2, \dots, y_m) = (y_1 - k_1, y_2 - k_2, \dots, y_m - k_m)$$
- All above operations are performed in Z_{26}

- Plaintext : “thisscryptosystemisnotsecure”.
- Encryption: add modulo 26

19	7	8	18	2	17	24	15	19	14	18	24	18	19	4	12
2	8	15	7	4	17	2	8	15	7	4	17	2	8	15	7
21	15	23	25	6	8	0	23	8	21	22	15	20	1	19	19

8	18	13	14	19	18	4	2	20	17	4
4	17	2	8	15	7	4	17	2	8	15
12	9	15	22	8	25	8	19	22	25	19

- Ciphertext:
“VPXZGIAXIVWPUBTTMJPWIZITWZT”.

Transposition/Permutation Cipher

- Let m be a positive integer
- $\mathcal{P} = \mathcal{C} = (Z_{26})^m$
- $\mathcal{K}$ = set of all possible permutations of $\{1, 2, \dots, m\}$
- For each permutation $\pi \in \mathcal{K}$,
$$e_{\pi}(x_1, x_2, \dots, x_m) = (x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(m)})$$
$$d_{\pi}(y_1, y_2, \dots, y_m) = (y_{\pi^{-1}(1)}, y_{\pi^{-1}(2)}, \dots, y_{\pi^{-1}(m)})$$
- π^{-1} being the inverse permutation of π

Example

- $m = 6$.
- key is the following permutation π :

x	1	2	3	4	5	6
$\pi(x)$	3	5	1	6	4	2

- inverse permutation π^{-1} :

x	1	2	3	4	5	6
$\pi^{-1}(x)$	3	6	1	5	2	4

- Plaintext : “defendthehilltopatsunset”

- partition the plaintext into group of six letters:

defend | thehil | ltopat | sunset

- rearrange according to π :

fnddee | eitlhh | oaltpt₆ | nestsu

- Ciphertext: “FNDDEEEITLHHOALTPTNESTSU”
- Decryption can be done using π^{-1}

Playfair Cipher

- Use the keyword CHARLES (Charles Wheatstone invented the cipher).
- Draw up a 5×5 matrix with the keyword first, removing any repeating letters as follows:

c	h	a	r	l
e	s	b	d	f
g	i/j	k	m	n
o	p	q	t	u
v	w	x	y	z

- Plaintext: “meet me at the bridge”.
- Split the sentence into digrams removing spaces, ‘x’ used to make even number of letters:

me et me at th eb ri dg ex
- Repeating plaintext letters that are in the same pair are separated with a filler letter, such as x:

“balloon” would be treated as ba lx lo on

- Two plaintext letters in the same row are each replaced by the letter to the right, with the first element of the row circularly following the last.

eb is replaced by sd

ng is replaced by gi (or gj as preferred)

Two plaintext letters that fall in the same column are each replaced by the letter beneath, with the top element of the column circularly following the last.

dt would be replaced by my

ty would be replaced by yr

- Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter.

me becomes gd

- Ciphertext therefore is :

“gd do gd rq pr sd hm em bv”

